Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[12pt, twoside]book [T1]fontenc [utf8]inputenc [english, spanish, es-noshorthands]babel [mono=false]libertine [notext]stix

PLANTILLA PARA TRABAJO DE FIN DE GRADO O MÁSTER U. N. Alumno 19 Pebruari 2025

tikz USredcmyk0,1.00,0.65,0.34

[]tabularx []enumitem noitemsep

[]amsmath []amsthm []mathtools []bm []thmtools [spaceabove=acebelow=headfont=USred, **notefont=USred, notebraces=(), bodyfont=**, *postheadspace=1em, numberwithin=chapter, headindent=0pt, headpunct=.]importante [spaceabove=acebelow=headfont=USred, notefont=*, notebraces=(), bodyfont=, postheadspace=1em, numberwithin=chapter, headindent=0pt, headpunct=.]normal [spaceabove=acebelow=headfont=*USred, notefont=*, notebraces=(), bodyfont=, postheadspace=1em, headindent=0pt, headpunct=., numbered=no, qed=USred]demostracion

[name=Observación, style=normal]remark [name=Corolario, style=normal]corollary [name=Proposición, style=normal]proposition [name=Lema, style=normal]lemma

[name=Teorema, style=importante]theorem [name=Definición, style=importante]definition

[name=Demostración, style=demostracion]proof

[a4paper, textwidth=80ex,]geometry

[explicit]titlesec

[hang] 2 USred 0pt 1

2.0 1ex 1

2.0.0 1ex 1

[]fancyhdr []emptypage plain tfg [LE]USred**7** [RO] **USred7**


sello.pdf



Memoria presentada como parte de los requisitos para la obtención del título de Grado en Matemáticas por la Universidad de Sevilla.

Tutorizada por

8.45cm[b]@l Prof. Tutor Primero Apellido Apellido
Prof. Tutor Segundo Apellido Apellido
Prof. Tutor Tercero Apellido Apellido

Daftar Isi

1 PENDAHULUAN **3**

2 TINJAUAN PUSTAKA **5**

English Abstract **1**

3 Los enunciados **3**

3.1 Teoremas y demostraciones 3

3.1.1 Otros enunciados 3

English Abstract

According to the guidelines, every dissertation should include a short english abstract at the beginning. In the abstract, you describe in general terms what is your dissertation about, the main points you want to make, and any important consequences that may arise.

Bab 3

Los enunciados

3.1 Teoremas y demostraciones

[Euclides] Esto es un Teorema. Se numeran a partir del 1 en cada capítulo. Como son importantes, tienen un cuadrado rojo al principio. Llevan letra cursiva.

Esto es la demostración. Al final de la demostración se puede ver un cuadrado rojo similar al de los teoremas. Las demostraciones no llevan letra cursiva.

Esto es una definición. Las definiciones son importantes; también llevan un cuadradito rojo.

3.1.1 Otros enunciados

Esto es una observación, que dice que $e = mc^2$. Como las observaciones no son importantes, no llevan cuadrado rojo, y el tipo de letra no es cursiva.

Si la demostración acaba en una fórmula, para poner el cuadrado rojo a la altura de la última formula, hay que usar la orden `\qedhere`, como en este caso:

$$e = mc^2.$$

Esto es un corolario.

Esto es una proposición.

[Gauss] Esto es un lema.